

## Part 2: EMU8086 Emulator & Operations

1. **How do you install EMU8086 on Windows?** Download the installer (setup.exe), run it, and follow the simple on-screen "Next" prompts. It installs like any standard Windows program.
2. **What are the system requirements for EMU8086?** It is very lightweight. It runs on almost any Windows version (XP to 11) and requires minimal RAM and disk space.
3. **How do you configure DOSBox for assembly programming?** EMU8086 has an *internal* emulator, so no config is needed. If using external DOSBox, you use the `mount` command to link a folder on your PC to a virtual drive letter (like C:).
4. **What is the difference between MASM and EMU8086?** MASM is a professional assembler by Microsoft. EMU8086 is an educational tool that combines an editor, assembler, and visual debugger all in one easy interface.
5. **How do you create a new assembly program in EMU8086?** Open EMU8086, click the "New" button, and select a template (like "COM template" or "EXE template").
6. **What file extension is used for 8086 assembly programs?** The source code files are saved with the `.asm` extension.
7. **How do you assemble and run a program in EMU8086?** Click the "Emulate" button to assemble the code. A new window pops up; click "Run" in that window to execute it.
8. **What is the difference between assemble and compile in EMU8086?** "Assemble" translates assembly code directly to machine code. "Compile" usually refers to saving the final standalone executable file (`.exe` or `.com`).
9. **How do you enable keyboard interrupts in EMU8086?** You don't need a specific setting. In your code, you simply call interrupt `INT 21h` with Function `01h` (read char) or `0Ah` (read string).
10. **How do you debug a program in EMU8086?** Click "Emulate," then use the "Single Step" button. This executes one line at a time so you can watch registers change value.
11. **How do you set memory segment size in EMU8086?** This is usually handled automatically by the templates (like `#MAKE_COM#`). You don't typically set this manually for basic exams.
12. **What is the difference between real mode and protected mode in DOS? Real Mode** (8086 default) gives direct access to hardware but has limited memory (1MB). **Protected Mode** prevents direct hardware access to stop programs from crashing the system.
13. **How do you save an assembled program in EMU8086?** In the main editor, click "Save". To create a runnable program file, click "Compile" to generate a `.com` or `.exe` file.

14. **How do you open an existing program in EMU8086?** Click **File > Open** in the menu bar and select your `.asm` file.
15. **How do you view registers and memory contents during execution?** When you click "Emulate," the emulator window opens. The registers (AX, BX, etc.) are listed on the left, and memory is shown in a grid on the right.